Wearable PCB System Documentation

This document provides a detailed explanation of a wearable health and safety monitoring system composed of two printed circuit boards (PCBs). The device is designed to be worn like a watch and is built using materials such as silicone, fabric, and recycled plastic to ensure comfort, flexibility, and sustainability.

1. System Overview

The system is divided into two main PCBs. The top PCB contains processing and communication components, while the bottom PCB contains sensors that directly interact with the user's body.

The device continuously collects physiological data, processes it, and transmits alerts and location information when necessary.

2. Top PCB (Outer Layer)

The top PCB acts as the main control and communication unit.

2.1 Microcontroller (Arduino)

The microcontroller serves as the brain of the system. It runs programmed instructions, processes sensor data, and makes decisions based on predefined conditions.

2.2 SIM Module

The SIM module enables cellular communication. It allows the device to send messages, make calls, and transmit data over mobile networks.

2.3 GPS Module

The GPS module determines the device's location using satellite signals. It provides latitude and longitude coordinates used during emergency alerts.

2.4 Push Button

The push button allows the user to manually trigger an emergency alert. It sends a digital signal to the microcontroller when pressed.

2.5 Supporting Components

Additional components such as resistors, capacitors, and voltage regulators ensure stable and safe operation of the circuit.

3. Bottom PCB (Sensor Layer)

The bottom PCB is positioned against the user's skin and is responsible for collecting physiological data.

3.1 Temperature Sensor

This sensor measures body temperature and detects abnormal conditions such as fever.

3.2 Motion Sensor

The motion sensor detects movement and acceleration. It is used for activity monitoring and fall detection.

3.3 Pulse Oximeter

The pulse oximeter measures blood oxygen levels and heart rate using light absorption techniques.

3.4 Interconnections

Connectors between the two PCBs allow data and power transfer between sensors and the microcontroller.

4. System Operation Flow

Step 1: Sensors collect physiological data.

Step 2: Data is sent to the microcontroller.

Step 3: The microcontroller processes and evaluates the data.

Step 4: If abnormal conditions are detected or the button is pressed, an alert is triggered.

Step 5: The SIM module sends alerts, and the GPS module provides location data.

5. Two-PCB Design Rationale

Separating the system into two PCBs improves comfort, measurement accuracy, signal quality, and modularity. The bottom PCB ensures proper skin contact, while the top PCB houses more complex electronics.

6. Wearable Design and Materials

The device is designed to be worn like a watch on the wrist. The enclosure and straps are made from silicone, fabric, and recycled plastic to ensure durability, flexibility, comfort, and environmental

sustainability.

7. System Summary

This wearable system integrates sensing, processing, and communication technologies to monitor health conditions and provide emergency assistance when needed.